

Galactic Rotation from the Compact Fiber II: Structural Regime Split, Realized Coherence, and a Zero-Parameter Benchmark Improvement

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Abstract

We present a second-stage compact-fiber model for galactic rotation curves that extends a previously published one-law formulation by introducing a restricted structural regime split in realized galactic organization. In the nominal regime, the effective coherence length remains the compact-fiber galactic scale L_{GL} . In the realized-transfer regime, the effective coherence length is given by a harmonic law combining L_{GL} with the baryonic half-mass radius $R_{1/2}$. Regime assignment is determined by a fixed structural proxy derived from post-core baryonic rotational organization, rather than by galaxy-by-galaxy fitting.

On the frozen 171-galaxy SPARC benchmark, the regime-split model improves on both the frozen one-law compact-fiber baseline and a fixed MOND comparator. The corresponding median χ^2/pt values are 10.71 for the one-law compact-fiber baseline, 10.05 for the regime-split compact-fiber model, and 11.25 for the MOND benchmark. The strongest class-level gain occurs in the giant systems, where the median improves from 47.34 to 29.11. Robustness checks show that the regime split is not a knife-edge threshold effect, that the harmonic transfer family is strongly preferred over nearby alternatives, and that the giant-class gain is stable across threshold perturbations.

The model also exhibits a known residual trade-off: a subset of realized-transfer-assigned moderate spirals is overcorrected in the mid-to-outer disk. A targeted refinement pass was therefore carried out, testing capped harmonic transfer, mild damping, and concentration-linked moderation. None of these simple moderations improved the spiral failure cluster without destroying the giant-class gain. The present regime-split harmonic model is therefore treated as the frozen result. We interpret this outcome as evidence that the regime split is a genuine structural extension of the one-law compact-fiber model, and that the remaining trade-off is structural rather than an artifact of threshold choice or simple over-tuning.

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1 Introduction

Galactic rotation curves remain one of the clearest galaxy-scale settings in which baryonic structure and dynamical response exhibit systematic regularities beyond naive Newtonian expectations. At this scale, Modified Newtonian Dynamics (MOND) continues to function as a serious benchmark because it provides a compact low-parameter description of those regularities [1, 2]. At the same time, standard halo-based modeling remains the mainstream framework, while alternatives such as self-interacting dark matter continue to be explored [3]. This leaves open a useful scientific niche for structurally motivated, low-parameter models that can be tested directly against galaxy-scale benchmark data.

The previous compact-fiber galactic-rotation paper introduced such a model: a one-law compact-fiber formulation built from compact-fiber-derived characteristic scales and applied uniformly across the SPARC benchmark sample [4]. That first formulation already produced a competitive galaxy-scale result without calibrated galaxy-by-galaxy freedom. However, inspection of the frozen benchmark outputs showed that the single nominal coherence realization law did not perform equally across the full sample. The strongest residual problems were not randomly distributed, but concentrated in particular parts of the galaxy population, especially among larger rotating systems with clearer retained baryonic structure.

The present paper tests a restricted structural extension motivated by that failure pattern. Its central hypothesis is that a single nominal compact-fiber coherence realization law is not adequate across the full benchmark sample, and that the sample is better described by two structural regimes of realized galactic organization. In one regime, the nominal compact-fiber galactic coherence scale remains sufficient. In the other, the effective realized coherence length is given by a harmonic transfer law combining the nominal compact-fiber coherence scale with the baryonic half-mass radius. Regime assignment is determined by a fixed structural proxy derived from post-core baryonic rotational organization. The point of this extension is not to introduce an open-ended family of class-conditional fits, but to test whether a restricted structural correction improves the frozen galaxy-scale benchmark.

The empirical outcome is encouraging but bounded. On the frozen 171-galaxy SPARC comparison, the regime-split model improves on both the frozen one-law compact-fiber baseline and a fixed MOND comparator. The strongest class-level signal is the giant-system improvement, where the median benchmark error drops sharply relative to the one-law baseline. At the same time, the model introduces a coherent residual weakness: a subset of realized-transfer-assigned moderate spirals is overcorrected in the mid-to-outer disk. A targeted refinement pass shows that no simple moderation removes that spiral failure cluster without destroying the giant-class gain. The trade-off therefore appears structural rather than accidental.

Accordingly, the scope of the paper is narrow. It presents a galaxy-scale structural improvement over a standard MOND benchmark, not a global replacement program for dark matter or a broad theory of galaxy development. Its contribution is twofold: first, a regime-split compact-fiber model that improves the frozen benchmark while preserving the zero-parameter character

of the comparison; second, evidence that the remaining spiral trade-off is a real boundary of the present formulation rather than a cheaply removable artifact.

The paper is organized as follows. Section 2 briefly recalls the compact-fiber ingredients needed for the galactic application. Section 3 introduces the regime-split hypothesis and the realized-transfer law. Section 4 defines the structural classifier and the frozen regime assignment. Section 5 describes the SPARC benchmark protocol, the authoritative frozen baseline, and the MOND comparator. Section 6 presents the main benchmark results, including the radial-acceleration-relation analysis. Section 7 reports robustness checks on threshold choice and transfer-family variants. Section 8 analyzes the realized-transfer spiral failure cluster, and Section 9 records the negative refinement result that justifies freezing the current model. Section 10 discusses the significance and scope of the result, and Section 11 concludes.

2 Compact-Fiber Background

The present paper is a direct continuation of the compact-fiber line developed in the preceding foundational and application papers [5, 6, 7, 4]. Only a minimal amount of background is needed here. The aim of this section is to state the specific compact-fiber ingredients required for the galactic-rotation application, not to re-derive the broader framework.

In the compact-fiber approach, the starting point is not a phenomenological galactic fit law but a compact-fiber carrier structure from which later effective laws are built [5]. For the galactic application, two derived scales are central. The first is the characteristic compact-fiber galactic acceleration scale

$$a_0 = a_{\text{nat}} \frac{N_m}{R^{16}}.$$

The second is the nominal galactic coherence length

$$L_{\text{GL}} = s_{\text{nat}} \frac{R^{12}}{\text{kpc}}.$$

In the present line, both are treated as derived quantities rather than tuned astrophysical fit parameters [4].

The original one-law galactic-rotation model used these scales uniformly across the full benchmark sample [4]. In that formulation, the total acceleration was written as

$$g(r) = g_{\text{bar}}(r) + \sqrt{a_0 \tilde{g}(r)},$$

where $g_{\text{bar}}(r)$ is the Newtonian baryonic acceleration from the observed gas, disk, and bulge components, and $\tilde{g}(r)$ is a nonlocal smoothed baryonic field. The compact-fiber coherence length enters through this smoothed field, with the effective realization length taken uniformly as the nominal scale L_{GL} .

That one-law formulation is the frozen baseline against which the present paper is measured. The compact-fiber foundation itself is not changed here. The same derived scales are retained, as is the overall structural form of the galaxy-scale acceleration law. What changes is the hypothesis about how coherence is realized across the benchmark sample. The question is no longer whether the compact-fiber line can generate a competitive galaxy-scale law at all; the earlier paper already showed that it can [4]. The question addressed here is whether a single nominal coherence realization is adequate across all systems.

The present paper argues that it is not. Instead, the benchmark sample appears to require a distinction between a regime in which the nominal coherence scale remains adequate and a regime in which the effective realized coherence depends additionally on retained baryonic organization. For that reason, the present model should be read as a structural extension of the one-law compact-fiber galactic model rather than as a new independent theory.

3 From One-Law Failure to Structural Regime Split

The present paper begins from a specific empirical observation: the frozen one-law compact-fiber model does not fail uniformly across the SPARC benchmark sample. Its strongest mismatches are concentrated in particular parts of the sample rather than distributed like random noise. This motivates the central hypothesis of the paper: the benchmark is better described by a restricted *structural regime split* than by a single universal coherence-realization law.

The compact-fiber foundation itself remains fixed, as do the derived scales a_0 and L_{GL} . What changes is the effective realization of the galactic coherence scale. In some systems, the nominal compact-fiber realization appears sufficient. In others, the effective realized coherence appears to depend not only on the nominal compact-fiber scale but also on a retained baryonic organization scale. The model therefore distinguishes between two structural regimes of realized galactic organization.

In the *nominal regime*, the effective coherence length is simply

$$L_{\text{eff}} = L_{\text{GL}}.$$

This is the direct continuation of the one-law compact-fiber formulation.

In the *realized-transfer regime*, the effective coherence length is taken to be

$$L_{\text{eff}} = \frac{2L_{\text{GL}}R_{1/2}}{L_{\text{GL}} + R_{1/2}},$$

where $R_{1/2}$ is the baryonic half-mass radius. This is a harmonic transfer law between the nominal compact-fiber coherence scale and a retained baryonic core scale. The motivation for this form is not that it is a generic interpolating function, but that it provides a low-complexity realization in which neither scale is discarded.

This realized-transfer law is intended as a hypothesis about how galactic coherence is effectively realized, not as a free deformation introduced separately for different galaxy classes. It is one fixed transfer law applied to one fixed structural regime, with no galaxy-by-galaxy calibrated freedom in the frozen benchmark comparison. In that sense, the present model is a restricted structural extension of the one-law compact-fiber baseline rather than a move into unrestricted conditional fitting.

The structural interpretation of the two regimes is straightforward. In the nominal regime, the compact-fiber coherence scale is effectively realized without further modification. In the realized-transfer regime, the observed galactic organization indicates that the nominal coherence scale alone is no longer the best effective descriptor, and that coherence is instead realized through a transfer relation involving both the nominal scale and the retained baryonic core scale. This is the full content of the regime-split hypothesis as used in the present paper.

The harmonic transfer form is itself part of the claim. As the robustness analysis will show, the harmonic family is strongly preferred over nearby geometric, arithmetic, and minimum-law alternatives. Within the tested family, it is the only low-complexity transfer form that preserves the strongest benchmark gains, especially in the giant systems. This is one of the main reasons the present regime-split model can be treated as a meaningful structural result rather than as an arbitrary empirical patch.

The regime-split hypothesis therefore has two components. First, the benchmark sample is better described by two structural coherence-realization regimes than by one universal nominal law. Second, the realized-transfer regime is governed by a fixed harmonic law connecting the nominal compact-fiber coherence scale and the retained baryonic core scale. The next section specifies the structural proxy used to assign galaxies to these two regimes in the frozen benchmark model.

4 Classifier and Regime Assignment

The regime-split model requires a fixed rule for assigning each galaxy to either the nominal regime or the realized-transfer regime. In the present paper, that assignment is made through a structural proxy derived from the baryonic rotation field outside the retained core region. Its role is to distinguish systems whose post-core baryonic organization is consistent with the nominal coherence realization from those for which the realized-transfer law appears to be required.

The proxy used in the frozen benchmark is

$$S_{\text{post}} = \text{median}_{r \geq R_{1/2}} \frac{d \ln v_{\text{bar}}}{d \ln r},$$

where $v_{\text{bar}}(r)$ is the baryonic circular-velocity contribution and $R_{1/2}$ is the baryonic half-mass radius. The subscript “post” denotes *post-core*: the statistic summarizes the typical logarithmic slope of the baryonic rotation field outside the retained core scale. It is therefore intended to capture a coarse feature of post-core baryonic organization rather than a local fluctuation at any one radius.

The realized-transfer assignment adopted in the frozen benchmark is

$$S_{\text{post}} \leq -0.02.$$

Galaxies satisfying this inequality are assigned to the realized-transfer regime; all others remain in the nominal regime. The threshold is fixed in the frozen benchmark and is not chosen separately for different galaxy classes. As the robustness results will show, the benchmark performance is not concentrated at a knife-edge threshold, but instead persists across a broad neighborhood of nearby threshold values.

The use of S_{post} has a specific rationale. The one-law compact-fiber model already suggested that the main mismatches were associated not with the inner baryonic core itself, but with how the baryonic rotational organization continued beyond that core. A classifier based on post-core baryonic slope therefore addresses the part of the system most likely to reveal whether the nominal coherence realization remains adequate. In this sense, the classifier is structural: it asks whether the observed baryonic organization beyond the retained core region is shallow enough to be consistent with one realization law or steep enough to signal the need for another.

The present classifier should also be distinguished from a crude velocity cut. The benchmark sample contains both nominal- and realized-transfer-assigned systems across a range of velocity classes. The realized-transfer bucket is enriched in larger and more strongly organized rotators, but it is not identical to a simple high- V_{flat} selection. The regime assignment therefore carries additional structural information beyond a one-parameter class label.

For the frozen benchmark, the authoritative regime counts are

$$N_{\text{realized-transfer}} = 98, \quad N_{\text{nominal}} = 73.$$

These counts are taken from the frozen benchmark outputs and are the counts used throughout the present paper. Some later diagnostic files mixed updated classification diagnostics with previously frozen benchmark summaries; those mixed diagnostics are not treated as authoritative here. For the same reason, recomputed diagnostic scans may yield slightly different regime counts at nearby thresholds than the frozen benchmark output; the authoritative counts for the present paper remain the frozen 98/73 split.

The classifier is empirically motivated and robust in the benchmark, but it is not yet derived from the compact-fiber framework. A direct derivation would promote the regime split from an effective structural observation to a theoretical prediction. At present, the claim is narrower: a fixed structural classifier of this type supports a robust and informative regime split, and the resulting gains are not artifacts of a fragile threshold choice.

In summary, the regime assignment in the present model is determined by a fixed post-core baryonic organization proxy. Its function is to separate systems that remain adequately described by the nominal compact-fiber coherence realization from systems whose realized organization is better captured by the harmonic realized-transfer law. The classifier is therefore an essential part of the frozen benchmark model, but it remains subordinate to the main structural claim: the benchmark sample is better described by two coherence-realization regimes than by one universal nominal law.

5 Data, Benchmark Protocol, and Comparator Models

The empirical benchmark used in this paper is drawn from the SPARC Rotmod_LTG catalog of late-type galaxy rotation curves [8, 9]. The repository accompanying this paper includes the SPARC data archive used in the experiments and preserves the exact benchmark outputs on which the present comparisons are based. The purpose of this section is to state precisely the sample, metric, comparator models, and authoritative files used in the frozen analysis.

The underlying SPARC Rotmod_LTG set contains 175 galaxy files [8]. In the benchmark protocol used here, 171 galaxies pass the inclusion rule requiring a minimum of five usable radial data points. All benchmark medians, classwise summaries, and comparator counts reported in the present paper refer to this frozen 171-galaxy sample. The data consist of observed rotation curves together with baryonic component decompositions for gas, disk, and bulge, allowing the baryonic circular-velocity contribution $v_{\text{bar}}(r)$ and the associated Newtonian baryonic acceleration $g_{\text{bar}}(r)$ to be computed consistently across the sample [8].

The primary benchmark metric is the per-galaxy χ^2/pt , computed for each galaxy and then summarized by sample medians and by classwise medians. The headline benchmark numbers in this paper are therefore medians over per-galaxy χ^2/pt values, not pooled global χ^2 values across all radial points. This distinction matters because the paper compares how different low-parameter laws behave across a heterogeneous galaxy sample rather than minimizing one combined residual over the entire dataset. The same metric is used for the one-law compact-fiber baseline, the regime-split compact-fiber model, and the MOND comparator.

The authoritative frozen one-law compact-fiber baseline is the result stored in

`results/frozen_baseline/baseline_results.csv`.

This file is generated by

`experiments/run_frozen_baseline.py`,

which implements the one-law compact-fiber galactic model. The authoritative frozen regime-split result is stored in

`results/regime_split/regime_split_results.csv`,

generated by

`experiments/run_regime_split.py`.

These two files are the authoritative sources for the main benchmark comparison in the present paper. Diagnostic files generated during later robustness work are not treated as authoritative unless they reproduce or summarize these frozen results explicitly.

A full reproducibility map linking the reported benchmark numbers, tables, and figures to their generating scripts and result files is provided in the companion repository.

The MOND comparator is a fixed benchmark law applied under the same dataset and metric conventions. In the code path used for comparison, the total MOND acceleration is written as

$$g_{\text{MOND}}(r) = \frac{g_{\text{bar}}(r)}{1 - \exp(-\sqrt{g_{\text{bar}}(r)/a_{0,\text{MOND}}})},$$

with

$$a_{0,\text{MOND}} = 1.2 \times 10^{-10} \text{ m s}^{-2}.$$

This is the exponential interpolation form used as the fixed MOND benchmark in the present comparison [1, 2]. In the frozen benchmark protocol, the MOND comparator is evaluated under the same fixed dataset and metric conventions as the compact-fiber models; it is not a per-galaxy free- M/L optimization exercise. The purpose of the comparator is therefore narrow: to provide a standard low-parameter galaxy-scale benchmark under the same protocol.

The benchmark protocol is therefore defined by four ingredients: the frozen 171-galaxy SPARC sample, the per-galaxy χ^2/pt metric summarized by medians, the frozen one-law compact-fiber baseline results file, and the frozen regime-split compact-fiber results file, together with the fixed MOND comparator evaluated under the same metric. This is the basis on which all headline comparisons in the present paper are made.

In addition to the frozen benchmark files, the repository contains supporting outputs for threshold scans, proxy-family variants, failure-cluster diagnostics, and radial-acceleration-relation analysis. These are used in later sections to assess robustness and structural limitations of the regime-split model. They do not replace the frozen benchmark comparison itself. The frozen baseline and frozen regime-split result files remain the authoritative reference points for the main claims of the paper.

6 Results

This section presents the frozen benchmark comparison between the one-law compact-fiber baseline, the regime-split compact-fiber model, and the fixed MOND comparator. All headline values reported here are taken from the authoritative frozen result files described in Section 5. The purpose of this section is to state the main empirical outcome of the paper before turning to robustness and failure-cluster analysis.

The primary benchmark comparison is summarized by the sample median of per-galaxy χ^2/pt over the 171-galaxy SPARC sample. The frozen headline values are

$$\text{median}(\chi^2/\text{pt})_{\text{baseline}} = 10.7064,$$

$$\text{median}(\chi^2/\text{pt})_{\text{regime split}} = 10.0518,$$

and

$$\text{median}(\chi^2/\text{pt})_{\text{MOND}} = 11.2504.$$

Thus the regime-split model improves on both the frozen one-law compact-fiber baseline and the fixed MOND comparator. The global gain is modest; the structural signal is clearest at the class level, especially in the giant systems.

Table 1: Frozen benchmark comparison on the 171-galaxy SPARC sample. The regime-split compact-fiber model improves on both the frozen one-law compact-fiber baseline and the fixed MOND comparator.

Model	Parameters	Median χ^2/pt	Wins vs MOND
MOND benchmark	fixed benchmark law	11.2504	—
Compact-fiber one-law baseline	0 (frozen structural scales)	10.7064	86/171
Compact-fiber regime split	0 (frozen structural split)	10.0518	90/171

A complementary view is given by pairwise galaxy counts. On the frozen benchmark, the regime-split model improves on the one-law baseline for 42 galaxies, worsens for 56, and leaves 73 unchanged. It improves on the MOND comparator for 90/171 galaxies. These counts show

that the gain is not a universal per-galaxy improvement over the one-law baseline. Rather, the regime split alters a subset of the sample while leaving a substantial fraction unchanged. Among the galaxies that improve under the regime split, the median reduction in χ^2/pt is 2.37, whereas among the galaxies that worsen, the median increase is 2.03. The benchmark gain therefore reflects a structured redistribution rather than uniform per-galaxy improvement. This is why the later robustness and failure-cluster analysis are essential to the interpretation of the result.

The classwise medians clarify where the gain is strongest.

Table 2: Velocity-class medians for the frozen one-law baseline and the regime-split model. The strongest class-level gain occurs in the giant systems.

Class	Baseline median χ^2/pt	Regime-split median χ^2/pt
Dwarfs < 50	10.05	10.05
Mid-low 50–100	7.65	6.09
Spirals 100–250	12.93	12.30
Giants ≥ 250	47.34	29.11

The dwarf bin is essentially unchanged, the mid-low systems show a clear improvement, and the 100–250 km s^{-1} spiral bin improves modestly. The dominant class-level signal remains the giant-system improvement. The giant median falls from 47.34 to 29.11, making this the clearest evidence that the structural regime split is capturing an organization difference that the one-law model could not express.

A representative giant-system success is shown in Figure 1.

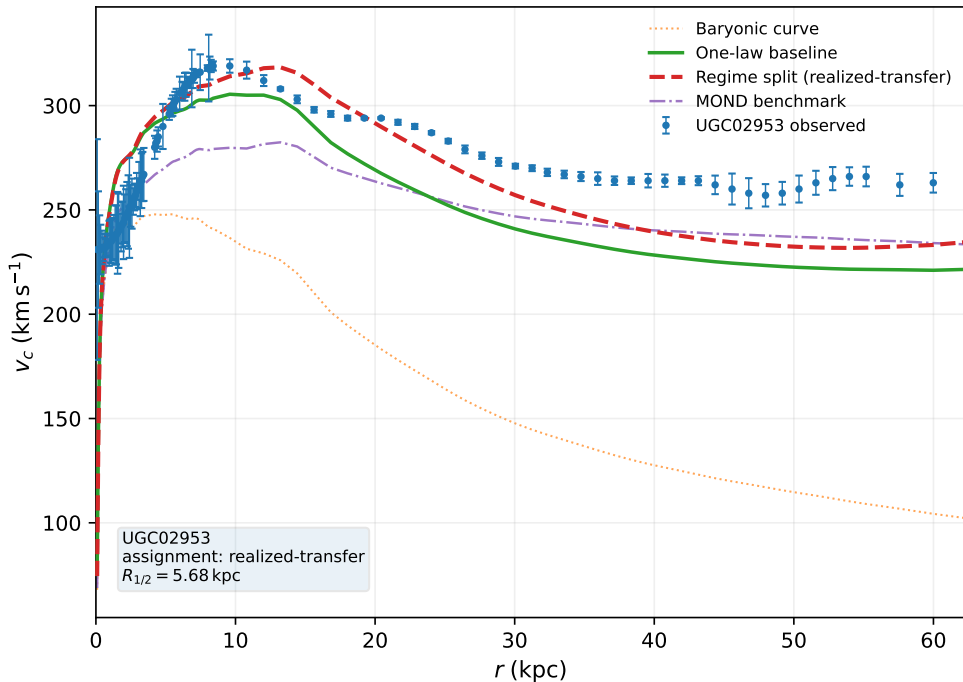


Figure 1: Representative giant-system success under the regime-split model: UGC02953. The one-law compact-fiber baseline underpredicts the outer rotation curve over a broad radial range, while the realized-transfer law tracks the observed high-velocity plateau substantially more closely. This illustrates the main empirical hook of the paper: the structural regime split produces its strongest and most robust gain in the giant systems.

The frozen regime counts are also informative. In the authoritative benchmark, the classifier

assigns

$$N_{\text{realized-transfer}} = 98, \quad N_{\text{nominal}} = 73.$$

This split does not reduce to a simple velocity taxonomy. The realized-transfer bucket is enriched in larger and more strongly organized rotators, but it is not identical to a crude high- V_{flat} cut. The regime assignment therefore carries additional structural information beyond a one-parameter class label.

It is also necessary to clarify the relation between the present frozen benchmark and earlier headline reporting. Prose associated with the earlier galactic-rotation paper reported a one-law compact-fiber headline value of 10.71 [4]. However, the frozen reproducible baseline CSV preserved in the benchmark workflow yields 10.7064. For the purposes of the present paper, the frozen reproducible value is treated as authoritative. The benchmark progression is therefore

$$10.7064 \rightarrow 10.0518$$

for the compact-fiber baseline and its regime-split extension, with the fixed MOND comparator at

$$11.2504.$$

All headline numbers in the present paper are taken from the frozen reproducible outputs.

6.1 Radial acceleration relation

In addition to the rotation-curve benchmark, the compact-fiber model naturally induces a radial acceleration relation between the baryonic acceleration g_{bar} and the predicted total acceleration g_{tot} . This is not imposed as a separate fit target. Rather, it emerges from the same gravitational-limit mechanism and nonlocal smoothing structure used in the rotation-curve calculation.

In the frozen analysis, the compact-fiber-predicted $g_{\text{bar}} \mapsto g_{\text{tot}}$ relation tracks the MOND benchmark interpolation closely over approximately four decades in acceleration, with deviations typically within ± 0.07 dex. The model-implied compact-fiber RAR scatter is approximately 0.06 dex, compared with an observed scatter of approximately 0.20 dex in the benchmark data. This is qualitatively consistent with the difference between a deterministic model relation and observational data containing measurement noise and galaxy-to-galaxy variation.

The regime split also produces a testable structural prediction in the RAR plane. At low accelerations, realized-transfer-assigned systems lie slightly above nominal-regime systems in the compact-fiber-predicted relation, with an offset of roughly 0.04 dex, while the two branches converge again at higher accelerations. The observed data show a qualitatively similar separation, but with a somewhat larger typical amplitude of about 0.08 dex in the low-acceleration bins where both branches are populated. In addition, the observed RAR scatter is lower for realized-transfer-assigned systems than for nominal-regime systems, consistent with the interpretation that the former have more coherent post-core baryonic rotational structure.

The main RAR-side systematic is also consistent with the rotation-curve residual analysis. At high accelerations, the compact-fiber prediction tends to lie about 0.07 dex above the MOND benchmark line. This corresponds to the same inner- and mid-disk overprediction pattern already seen in the radial residuals. Thus the RAR analysis does not reveal a new inconsistency; it recovers the same structural limitation in a complementary representation.

This behavior is shown in Figure 2.

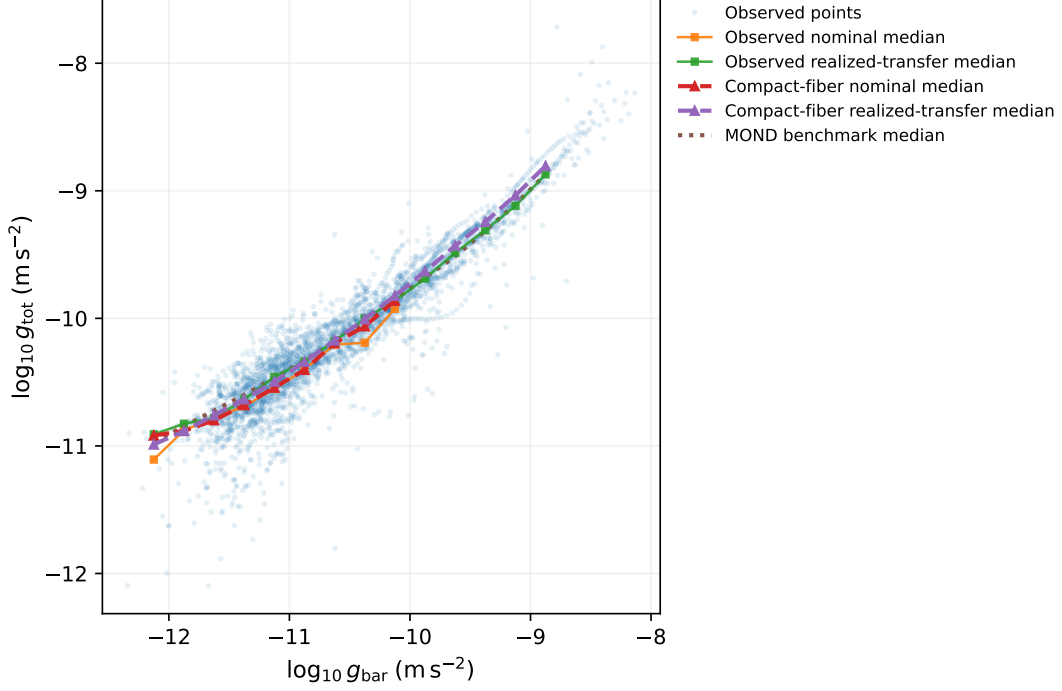


Figure 2: Radial acceleration relation for the frozen SPARC benchmark. Points show observed data, while binned medians separate the nominal and realized-transfer regimes. The compact-fiber model naturally generates an RAR-like relation without fitting to the RAR directly. The regime split predicts a small low-acceleration separation between the two structural regimes, while the high-acceleration offset reflects the same inner- and mid-disk overprediction seen in the rotation-curve residuals.

Taken together, the results support three immediate conclusions. First, the compact-fiber line remains competitive at galaxy scale without calibrated galaxy-by-galaxy freedom in the frozen comparison. Second, a restricted structural regime split improves the benchmark relative to the frozen one-law compact-fiber baseline. Third, the dominant empirical gain is concentrated in the giant systems, while the remaining weakness lies in a subset of realized-transfer-assigned moderate spirals. The next sections examine both points directly: Section 7 shows that the split is robust to threshold perturbations and that the harmonic transfer family is strongly preferred to nearby alternatives, while Section 8 identifies the main residual weakness of the model.

7 Robustness of the Regime Split

The benchmark improvement reported in Section 6 would be of limited value if it depended on a knife-edge classifier threshold or on an arbitrary transfer-law choice. For that reason, the regime-split model was subjected to a targeted robustness audit. The main results are straightforward: the classifier sits on a broad performance plateau, the harmonic transfer family is strongly preferred over nearby alternatives, and the giant-class gain remains stable under the tested perturbations.

We begin with threshold robustness. The frozen classifier assigns galaxies to the realized-transfer regime when

$$S_{\text{post}} \leq -0.02,$$

where S_{post} is the post-core baryonic organization proxy defined in Section 4. Recomputing the benchmark over a wide range of nearby threshold values shows that the global median does not concentrate at a single optimum. Across the tested range from -0.20 to $+0.20$,

the recomputed median remains within the narrow band 9.93 to 10.06, compared with the frozen one-law baseline median of 10.7064. The adopted threshold -0.02 gives the authoritative headline regime-split result of 10.0518, while nearby thresholds reach minima of 9.93. The regime-split gain is therefore not a threshold-fragile result. Rather, the threshold scan is best interpreted as a stability diagnostic showing a broad performance plateau rather than a knife-edge optimum.

Table 3: Threshold robustness scan for the realized-transfer assignment. The global median remains on a broad plateau, and the giant-class median remains fixed at 29.11 across the tested range. The adopted threshold is -0.02 , which yields the authoritative headline regime-split result of 10.0518; nearby thresholds reach minima of 9.93.

Threshold	Realized-transfer	Nominal	Median χ^2/pt	Giant median	Wins vs MOND
-0.15	78	93	9.93	29.11	89/171
-0.10	88	83	10.05	29.11	89/171
-0.08	91	80	10.05	29.11	89/171
-0.06	95	76	10.05	29.11	90/171
-0.04	97	74	10.05	29.11	90/171
-0.02	98	73	10.05	29.11	90/171
0.00	105	66	10.05	29.11	90/171
$+0.02$	111	60	10.06	29.11	90/171
$+0.05$	119	52	10.06	29.11	90/171
$+0.10$	128	43	9.93	29.11	90/171
$+0.15$	133	38	9.93	29.11	90/171
$+0.20$	137	34	9.93	29.11	90/171

The giant-class gain is even more stable. Across the same threshold scan, the giant-class median remains

$$29.11,$$

whereas the one-law baseline giant median is

$$47.34.$$

Within the tested threshold neighborhood, the strongest class-level success of the model is therefore unchanged. This is one of the main reasons the classifier can be treated as structurally informative rather than as a narrow tuning device.

The second robustness question concerns the transfer-law family itself. The frozen regime-split model uses the harmonic realized-transfer law

$$L_{\text{eff}} = \frac{2L_{\text{GL}}R_{1/2}}{L_{\text{GL}} + R_{1/2}}.$$

To test whether this was an arbitrary choice, several nearby alternatives were examined, including geometric-mean, arithmetic-mean, and minimum-law variants, together with harmonic variants using different retained-mass fractions such as $R_{1/3}$, $R_{1/2}$, $R_{2/3}$, and $R_{3/4}$.

The outcome is clear. Geometric and arithmetic mean laws degrade the global benchmark median, and the minimum-law variant largely removes the giant-class gain. By contrast, the harmonic family consistently preserves the main structure of the result. Within that family, the exact retained-core fraction is only weakly load-bearing at the level of the global median: harmonic variants based on $R_{1/3}$, $R_{1/2}$, $R_{2/3}$, and $R_{3/4}$ all remain in a narrow global-median band around 10.05–10.06, while the giant-class median improves smoothly as the retained fraction increases. The decisive structural choice is therefore the adoption of the harmonic transfer family itself, not fine selection of one retained fraction.

Table 4: Transfer-family robustness comparison. The harmonic family is strongly preferred over nearby alternatives. Within the harmonic family, the precise retained-core fraction is only weakly load-bearing at the level of the global median.

Variant	Median χ^2/pt	Giant median	Wins vs MOND
Baseline (no split)	10.71	47.34	86/171
$R_{1/3}$ harmonic	10.06	32.61	91/171
$R_{1/2}$ harmonic (adopted)	10.05	29.11	90/171
$R_{2/3}$ harmonic	10.05	27.33	90/171
$R_{3/4}$ harmonic	10.05	26.68	90/171
$R_{1/2}$ geometric mean	13.60	35.01	71/171
$R_{1/2}$ minimum	10.71	47.34	86/171
$R_{1/2}$ arithmetic mean	15.25	56.61	68/171

A further robustness check concerns how the benchmark gain is distributed across the sample. The improvement and degradation audit shows that the effect is not random. The strongest improvements are concentrated in high- V_{flat} realized-transfer-assigned systems, including both large spirals and giant galaxies. The strongest degradations are concentrated in a smaller subset of realized-transfer-assigned moderate spirals. This means that the regime split is not merely reshuffling errors diffusely across the sample. It is responding to a real structural division, albeit imperfectly.

The robustness audit therefore supports four conclusions. First, the regime split is not a knife-edge threshold effect. Second, the giant-class gain is highly stable. Third, the harmonic family is strongly preferred over nearby transfer-law alternatives. Fourth, the redistribution of benchmark error is structured rather than random. These points do not show that the model is complete, but they do show that its main strengths cannot be dismissed as artifacts of threshold tuning or casual law choice.

Because the threshold is robust and the harmonic family is strongly preferred, the main remaining weakness of the model becomes more informative. The unresolved issue is not whether the split is real, but why the same full-strength harmonic transfer that improves the giant systems also overcorrects a subset of realized-transfer-assigned moderate spirals. That failure cluster is the subject of Section 8.

8 Realized-Transfer Failure Cluster

The robustness audit of Section 7 shows that the regime split itself is not fragile and that the harmonic transfer family is strongly preferred over nearby alternatives. Once those points are established, the main remaining weakness of the model becomes much clearer. It is not a diffuse degradation spread randomly across the sample. Instead, it is concentrated in a specific subset of galaxies: realized-transfer-assigned moderate spirals for which the transfer law produces a systematic overcorrection in the mid-to-outer disk.

The leading degraded systems include UGC03546, NGC5055, UGC03580, and UGC05253. These galaxies are not marginal cases sitting near the assignment threshold. Their S_{post} values place them well inside the realized-transfer bucket. The failure cluster is therefore not a classifier-boundary artifact. The regime assignment remains structurally meaningful for these systems even though the realized-transfer law harms them.

The cluster also does not appear to be explained by a simple retained-core-scale separation. Aggregate comparisons between the degraded and improved realized-transfer-assigned systems show substantial overlap in $R_{1/2}$, in the ratio $R_{1/2}/L_{\text{GL}}$, in the classifier statistic S_{post} , and in the resulting realized-transfer scale L_{eff} . The degraded systems do, however, tend to occupy a

more moderate velocity range and often begin from less catastrophic one-law errors than the strongly improved giant systems. This already suggests that the issue is not the existence of the realized-transfer regime itself, but the detailed action of one transfer law within one part of that regime.

The most important diagnostic result is radial. When the degraded realized-transfer-assigned spirals are inspected in detail, the pattern is consistent. In the inner region, the one-law baseline and the regime-split model typically behave similarly. The realized-transfer law does not radically alter the inner prediction. The problem appears farther out. In the mid-to-outer disk, the regime-split residuals become systematically more positive than those of the one-law baseline, typically by about ten to fifteen percentage points in the inspected cases. Since positive residuals correspond here to predicted rotation speeds that are too large, the interpretation is straightforward: for this subset of moderate spirals, the harmonic realized-transfer law pushes the effective coherence realization too far into the outer disk. The result is a consistent outer-disk overcorrection.

A representative example of this failure mode is shown in Figure 3.

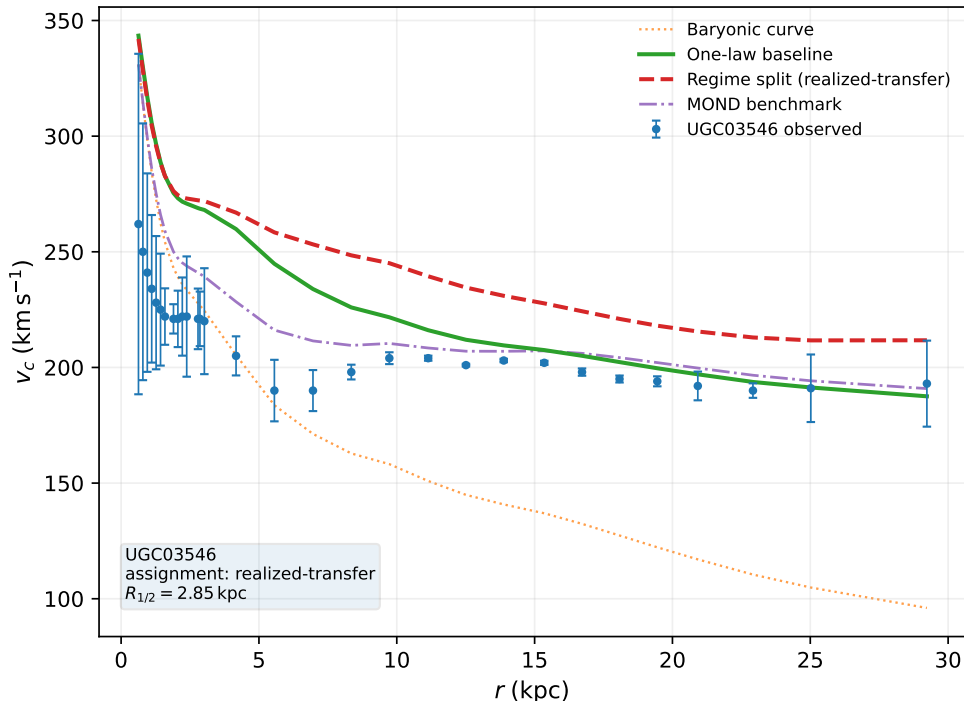


Figure 3: Representative degraded moderate spiral in the realized-transfer regime: UGC03546. The one-law baseline and regime-split predictions are similar in the inner region, but the regime-split law sits systematically higher in the mid-to-outer disk, producing the overcorrection characteristic of the realized-transfer spiral failure cluster. The MOND benchmark remains closer to the observed curve in the outer region, illustrating that this is a specific limitation of the present realized-transfer law rather than a generic difficulty for all low-parameter models.

This radial pattern matters because it identifies a specific failure mode. The model is not simply worse in some galaxies for an unclear reason. It is worse in a recognizable way: the full-strength harmonic realized-transfer law extends the nonlocal smoothing too effectively for certain moderate spirals, causing the predicted outer rotation curve to sit too high. This strongly supports the view that the residual problem is one of realized-transfer-law magnitude or shape, not of classifier existence. The realized-transfer bucket is still meaningful; what is imperfect is the use of one single transfer law for all systems in that bucket.

A further point of interpretation comes from comparator behavior. Some of the degraded

spirals are also difficult for the MOND benchmark, which suggests that they are intrinsically challenging cases for simple low-parameter galaxy laws. However, not all of them behave this way. A few systems are harmed much more strongly by the regime-split law than by MOND. The failure cluster therefore cannot simply be dismissed as a collection of universally pathological galaxies. At least part of the degradation appears to be specific to the present realized-transfer law rather than to galaxy modeling in general.

Taken together, these findings support three negative conclusions and one positive conclusion. First, the failure cluster is not a threshold-boundary problem. Second, it is not primarily explained by a simple separation in $R_{1/2}$ or related aggregate summary scales. Third, it is not just random residual noise. Positively, it is a coherent subset of moderate spirals for which the harmonic realized-transfer law is too strong in the outer disk.

The existence of this cluster does not negate the value of the regime split. On the contrary, it sharpens its meaning. The giant-class gain and the moderate-spiral degradation occur within the same broad realized-transfer regime, which means the transfer law itself is carrying real explanatory burden rather than merely reshuffling categories. The current model is therefore best understood as one that captures a genuine structural distinction in the benchmark sample, while still imposing too uniform a realized-transfer rule on the realized-transfer systems as a whole. The natural next question is whether that trade-off can be removed by a simple moderation of the transfer law. The answer, developed in the next section, is that within the tested simple refinement family it cannot.

9 Refinement Pass and Structural Trade-Off

Once the spiral failure cluster had been identified as a coherent outer-disk overcorrection within the realized-transfer regime, the natural next question was whether that weakness could be removed by a small refinement of the transfer law. The goal of the refinement pass was deliberately narrow. It was not to reopen unrestricted model exploration, nor to introduce a new family of galaxy-by-galaxy adjustments. Instead, it was to determine whether a simple, structurally interpretable moderation of the realized-transfer law could reduce the spiral degradation while preserving the main strengths of the regime-split model, especially the giant-class gain.

Three classes of refinement were tested. The first was a capped harmonic transfer, in which the harmonic realized-transfer law was retained but prevented from exceeding a chosen bound. The second was a mild damping of the harmonic transfer, in which the full-strength realized-transfer scale was reduced by a multiplicative moderation factor. The third was a concentration-linked moderation, in which the harmonic transfer was adjusted using a concentration-like structural measure in an attempt to distinguish the degraded spirals from the strongly improved giant systems.

The outcome of this refinement pass was unambiguous. None of the tested simple moderations improved the spiral failure cluster without materially degrading the giant-class gain. In the capped-harmonic family, caps weak enough to preserve the giant-class improvement were effectively inactive, while caps strong enough to help the degraded spirals immediately clipped the same mechanism responsible for the giant-class success. In the damped-harmonic family, any damping factor below full strength drove the giant-class median back toward its one-law baseline level. In the concentration-linked variants, the same pattern recurred: small moderations produced little useful spiral recovery, whereas stronger moderations damaged the giant systems more than they helped the degraded spirals.

This negative result is scientifically important. It shows that the remaining trade-off in the current model is not a loose artifact of a badly chosen threshold or an obviously excessive realized-transfer strength that can be trivially dialed back. Instead, within the tested family, the same full-strength harmonic realized-transfer law that produces the robust giant-class improvement is also responsible for the outer-disk overcorrection in a subset of moderate spi-

rals. The helpful and harmful effects are therefore coupled. That coupling is exactly what one would expect if the present model has reached a real structural boundary rather than merely an unoptimized parameter setting.

The interpretation is narrow but consequential. The refinement pass does not show that no better realized-transfer law can ever exist. What it shows is that, within the simple moderation family tested here, there is no clean path to improving the spiral failure cluster without sacrificing the strongest current success of the model. This is why the paper does not continue searching through a wider space of ad hoc modifications. Doing so would risk turning a structurally interpretable model into a patch system. The present aim is instead to freeze the strongest coherent result and describe its strengths and limits honestly.

Accordingly, the current regime-split harmonic formulation is taken as the frozen result of this paper. This decision is not an evasion of the residual weakness; it follows directly from the refinement evidence. The model is retained because it establishes three nontrivial points. First, the one-law compact-fiber baseline can be improved by a restricted structural regime split. Second, the harmonic transfer family is not an arbitrary choice but the only nearby family tested that preserves the main benchmark gains. Third, the remaining spiral overcorrection is not cheaply removable by simple moderation. That makes the trade-off itself part of the scientific result rather than an incidental defect.

The refinement pass therefore sharpens the interpretation of the paper in two ways. Positively, it increases confidence that the giant-class gain is a real feature of the structural extension rather than a fragile consequence of one arbitrary law choice. Negatively, it shows that the spiral failure cluster should be treated as a genuine limitation of the present formulation. The paper therefore ends not with a patched best fit, but with a constrained model whose main success and main limitation both arise from the same realized-transfer mechanism. That is the appropriate stopping point for the present stage of the compact-fiber galactic-rotation line.

10 Discussion

The results of this paper should be interpreted at a specific level of scope. The contribution is a galaxy-scale structural improvement over a standard MOND benchmark, not a global replacement program for all dark-matter or modified-gravity questions. MOND remains a meaningful comparator at this scale because it provides a compact low-parameter description of rotation-curve regularities even where its broader status remains contested [1, 2]. On that benchmark, the regime-split compact-fiber model improves the frozen median without introducing galaxy-by-galaxy calibrated freedom.

A further astronomy-facing result is that the compact-fiber model naturally generates an RAR-like relation without being fit to the RAR as a separate target. This matters because the RAR is often treated as the empirical regularity behind MOND-style galaxy-scale success. In the present case, the compact-fiber mapping from g_{bar} to g_{tot} tracks the MOND benchmark closely over a wide acceleration range, while the regime split adds a small low-acceleration branch separation between the two structural regimes. The observed data show a qualitatively similar separation, with lower scatter in the realized-transfer-assigned systems. This does not replace a full dedicated RAR analysis, but it does show that the regime-split compact-fiber model is engaging the same empirical regularity rather than merely reproducing a benchmark score.

The main conceptual advance over the earlier one-law compact-fiber model is not simply the lower global median. It is the introduction of a restricted structural regime split that addresses a patterned failure of the one-law formulation. The strongest empirical signal is the giant-system improvement. The global median shift from 10.7064 to 10.0518 is modest, but the giant-class median drops from 47.34 to 29.11, and that gain is stable across threshold perturbations and across the tested harmonic transfer family. This is where the main galaxy-scale scientific signal

of the paper resides.

At the same time, the model has a coherent limitation. A subset of realized-transfer-assigned moderate spirals is overcorrected in the mid-to-outer disk. The failure-cluster analysis shows that this is not a threshold-boundary artifact and not a simple retained-core-scale separation problem. The targeted refinement pass then shows that no simple moderation of the realized-transfer law removes that cluster without destroying the giant-class gain. This is one of the most important outcomes of the paper. It means that the present result is not a loose fit device that can be cheaply improved by trivial moderation. The trade-off is structural within the tested family.

Several limitations remain open. First, the classifier, though structurally interpretable and robust in the benchmark, is not yet derived from the compact-fiber framework. A direct derivation would promote the regime split from an effective structural observation to a theoretical prediction. Second, the realized-transfer law is supported empirically and comparatively within the tested family, but a stronger direct compact-fiber derivation would strengthen the line further. Third, the failure cluster suggests that the present realized-transfer law is still too uniform for all realized-transfer-assigned systems. The negative refinement pass shows that this cannot be fixed cheaply within the simplest moderation family, but it does not prove that no stronger structural refinement exists.

The relation to other alternatives should also be stated carefully. Standard dark-matter halo models remain the mainstream approach because they offer wide flexibility and broad cross-regime reach, though often at the cost of additional fit structure. MOND remains an important galaxy-scale benchmark because it captures key rotation-curve regularities compactly, though its broader limitations are well known [2]. Alternatives such as self-interacting dark matter illustrate the continuing search for structurally richer dark-sector explanations at astrophysical scales [3]. The present compact-fiber regime-split model occupies a different niche. It is a structurally motivated, low-parameter galaxy-scale model that improves on a standard MOND benchmark in the frozen SPARC comparison while exposing a real structural trade-off of its own.

The most appropriate overall interpretation is therefore conservative but positive. The paper does not claim a universal solution to galaxy dynamics. It does claim that a restricted compact-fiber structural extension improves a frozen one-law baseline and a standard MOND benchmark at galaxy scale, with a robust giant-class gain and a clearly identified residual limitation. That is the right level of claim for the present evidence.

11 Conclusion

This paper has presented a restricted structural extension of the one-law compact-fiber galactic-rotation model. The central question was whether a single nominal coherence realization law was adequate across the full SPARC benchmark sample. The present results indicate that it is not.

The main positive result is that a structural regime split in realized galactic organization improves the frozen benchmark. In the nominal regime, the effective coherence length remains the compact-fiber galactic scale L_{GL} . In the realized-transfer regime, the effective coherence length is given by a harmonic law combining L_{GL} with the baryonic half-mass radius $R_{1/2}$. On the frozen 171-galaxy SPARC comparison, this regime-split model improves the median χ^2/pt from 10.7064 for the one-law compact-fiber baseline to 10.0518, while the fixed MOND comparator remains at 11.2504. The strongest class-level gain occurs in the giant systems, where the median improves from 47.34 to 29.11.

The robustness analysis strengthens this result in three ways. The regime split does not depend on a knife-edge threshold choice. The harmonic realized-transfer family is strongly preferred over nearby geometric, arithmetic, and minimum-law alternatives. And the giant-

class gain remains stable across threshold perturbations and nearby harmonic retained-core variants. These findings support the interpretation that the present model is a genuine structural improvement over the frozen one-law baseline rather than an artifact of threshold tuning or casual law choice.

At the same time, the paper has identified a real residual limitation. A subset of realized-transfer-assigned moderate spirals is overcorrected in the mid-to-outer disk. Detailed inspection shows that this is not a threshold-boundary problem, and a targeted refinement pass shows that no simple moderation of the realized-transfer law removes this spiral failure cluster without destroying the giant-class gain. Within the family tested here, the same full-strength harmonic transfer that produces the strongest benchmark success also produces the remaining structured trade-off.

The appropriate conclusion is therefore bounded but positive. The present paper does not offer a universal solution to galaxy dynamics. It does show that, at galaxy scale and under the frozen SPARC benchmark protocol, a restricted compact-fiber regime split provides a better structural description than a single nominal coherence realization law, and does so while naturally generating an RAR-like relation. That is the main result of the paper. The next tasks for this line are to strengthen the direct derivational status of the classifier and realized-transfer law, and to determine whether the spiral trade-off points to a deeper subdivision of the realized-transfer regime or to a different retained-organization law altogether.

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